

Seyed Mahmood Shantiaeezade

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Summary

Robotics Ph.D. student at Oregon State University researching adaptive control and vibration suppression in mechatronic systems, with experience in modeling, embedded software, system identification, and experimental validation. Developed embedded motion-control platforms and contributed to Team OSU's first-place ASPE Student Challenge system. Brings a model-to-hardware approach to system integration, characterization, and root-cause troubleshooting of robotic subsystems.

Education

Ph.D. in Robotics

Sep. 2025 - Expected Sep. 2030.

Oregon State University, Corvallis, OR | GPA: 3.88 / 4.00

M.Sc. in Mechanical Engineering

Sep. 2023 - Sep. 2025

Oregon State University, Corvallis, OR | GPA: 3.92 / 4.00

B.Sc. in Mechanical Engineering

Sep. 2018 - Jul. 2023

University of Tehran, Tehran | GPA: 3.72 / 4.00

Research and Engineering Experience

Graduate Research Assistant | Sep. 2023 - Present

Manufacturing Process Control Laboratory

- Integrated spindle-synchronized motion commands with machine-tool X/Y feed drives and 5 kHz data acquisition to implement chatter suppression and increase the stable operating range using existing actuation hardware.
- Developed and experimentally validated an adaptive feedforward controller that adjusted motion amplitude and phase from measured force response, suppressing unstable vibration across multiple operating conditions without requiring an exact pre-identified process model.
- Developed kinematic and dynamic models of feed-modulated milling and validated stability predictions through physical experiments demonstrating an approximately **2-3x increase in the experimentally stable operating range** under tested conditions.
- Characterized feed-drive dynamics and nonlinear friction and implemented cascade feedback controllers and feedforward friction compensation in **Simulink** and **dSPACE** for machine-tool motion control.

Research Assistant | Feb. 2022 - Sep. 2023

Smart Electromechanical Energy Conversion Systems Laboratory

- Integrated magnetic position sensing, an STM32 microcontroller, and servo-amplifier circuitry to establish an experimental platform for voice-coil actuator characterization and embedded control.
- Designed and implemented **digital feedback control in C on STM32**, achieving a 10 mm positioning step in under 0.5 seconds with no observed overshoot and near-zero steady-state error.
- Identified actuator dynamics from chirp-response testing in **MATLAB**, achieving **90.75% model fit** to experimental data and using the model to design the positioning controller.
- Characterized syringe-pump friction through repeated measurements at each tested velocity and developed a nonlinear Simulink model to investigate discrepancies between predicted and measured positioning responses.

Research Intern | Jul. 2021 - Jan. 2022

Bio-MEMS and LOC Lab

- Developed a **C library for ATmega8** microcontroller to execute measurement routines for a coin-sized potentiostat.
- Designed and analyzed low-pass filtering circuitry and implemented Arduino measurement routines to support potentiostat development.

Selected Engineering Projects

Precision Motion Control for Musical Record Manufacturing | 2025

ASPE Student Challenge — Oregon State University, **First-Place Team Award**

- Implemented real-time motion control in **LabVIEW** on NI myRIO and collaborated on hardware integration, alignment, and experimental testing, contributing to the team's first-place finish.
- Characterized a voice-coil-driven flexure stage at five positions using sine-sweep testing and applied gain-scheduled inverse filtering to compensate for position-dependent dynamics.
- Designed a feedback controller with low-pass and notch filtering, achieving approximately 1 Hz tracking bandwidth with nominal **gain and phase margins of 17 dB and 83°**.

ArmPi Robotic Writing - Trajectory Generation and Experimental Integration

ROB 515 Course Project | Oregon State University | 2026

- Generated letter trajectories in **MATLAB**, discretized each path into 20 pen-up/pen-down waypoints, and implemented the motion sequence in **Python** on a **Raspberry Pi-controlled robot**.
- Integrated and calibrated the physical writing setup by stabilizing the marker and developing an iterative Python height-test routine for reliable paper contact.
- Troubleshoot vibration and path-execution errors by reducing waypoint density, increasing X–Y motion time to 1.8 seconds, and relocating the writing region closer to the robot base to reduce observed vibration during writing.

PiCar-X Autonomous Mobile Robot - Perception, Control, and Systems Integration

ROB 515 Course Project | Oregon State University | 2026

- Implemented and tested Python modules for camera and grayscale sensing, ultrasonic ranging, **steering-servo control, and DC-motor actuation**, including obstacle response at approximately 15 cm.
- Organized sensing, interpretation, and control using RossROS producer–consumer processes and inspected intermediate image, bus, and command outputs to isolate faults across the perception-to-actuation workflow.

Mechanical Robot Gripper Redesign

ME 512 Course Team Project | Oregon State University | 2025

- Redesigned and simulated a robot-gripper linkage in **Siemens NX**, evaluating end-effector kinematics, force transmission, mechanical advantage, and the effect of a two-stage gear train.

Technical Skills

Programming: MATLAB, Python, C, C++

Embedded and Real-Time Systems: STM32, Arduino, NI myRIO, dSPACE, Raspberry Pi

Control and Modeling: Simulink, LabVIEW, system identification, feedback and feedforward control, adaptive control, signal processing, frequency-response analysis

Mechatronics and Systems Integration: Sensor, motor, and actuator integration, embedded control, data acquisition, system characterization, test-method development, calibration, experimental testing, root-cause troubleshooting

Design and Electronics: SolidWorks, Siemens NX, COMSOL Multiphysics, NI Multisim, electrical circuit and schematic analysis

Selected Publications

- Bahtiyar, K., Bozcu, L., **Shantiaeezade, Seyed Mahmood**, & Sencer, B. "ADAPTIVE TOOL ECCENTRICITY COMPENSATION USING MACHINE TOOL FEED DRIVES." MM Science Journal (2025).
- Kawana, Y., **Shantiaeezade, Seyed Mahmood**, Sencer, B., Ikeda, R., & Suzuki, N., "Low frequency feed modulation assisted milling for chatter avoidance". CIRP Annals (2025).
- **Shantiaee Zade, Seyed Mahmood**, Robati, M., Nejad, M. G., Maleki, S., & Sadighi, A. (2022, November). "Optimal design and control of a planar voice-coil actuator for micropositioning applications". In 2022 10th RSI International Conference on Robotics and Mechatronics (ICRoM) (pp. 439-444). IEEE.

Teaching Experience

Teaching Assistant | Oregon State University | 2024 - 2025

Courses: Computer-Aided Design and Manufacturing; Computer Control of Manufacturing Processes; Lean Manufacturing Systems Engineering; Composite Manufacturing

Teaching Assistant | University of Tehran | 2021 - 2022

Courses: Automatic Control; Design of Machine Elements I